Robotics and Computer–Integrated Manufacturing 51 (2018) 222–229

Contents lists available at ScienceDirect

Robotics and Computer–Integrated Manufacturing

journal homepage: www.elsevier.com/locate/rcim

Review

Automated material handling in composite manufacturing using pick-and-place systems – a review

Andreas Björnsson[a,*], Marie Jonsson[b], Kerstin Johansen[c]

[a] Manufacturing Engineering, Linköping University, 58183 Linköping, Sweden
[b] Swerea Sicomp, Compraser Labs, Bröderna Ugglas Gata hus 208B, 58188 Linköping, Sweden
[c] Machine Design, Linköping University, 58183 Linköping, Sweden

ARTICLE INFO

Keywords:
Automation
Composite materials
Pick-and-place
Prepreg
Review

ABSTRACT

With increasing use of fiber reinforced polymer composites follows a natural pursuit for more rational and effective manufacturing. Robotic pick-and-place systems can be used to automate handling of a multitude of materials used in the manufacturing of composite parts. There are systems developed for automated layup of prepreg, dry fibers and thermoplastic blanks as well as to handle auxiliary materials used in manufacturing. The aim of this paper is to highlight the challenges associated with automated handling of these materials and to analyze the main design principles that have been employed for pick-and-place systems in terms of handling strategy, reconfigurability, gripping technology and distribution of gripping points etc. The review shows that it is hard to find generic solutions for automated material handling due to the great variety in material properties. Few cases of industrial applications in full-scale manufacturing could be identified.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

The use of Fiber Reinforced Polymers (FRP) grows steadily as the material properties present an opportunity to design strong, light structures with tailored properties. Passenger aircrafts like the Boeing 787 and Airbus A350 have increased the use of FRP in the aerospace sector and the introduction of electric and hybrid cars like the BMW i3 and i8 has dramatically increased use of FRP in automotive applications. For example the demand for carbon fiber is predicted to grow with more than 10% annually until 2020 [1,2].

In the aerospace sector FRP-products are commonly made by layup of thin sheets of prepreg (fibers pre-impregnated with polymer resin) either directly onto a mold or by layup of a flat laminate that is then subsequently formed. Manufacturing using prepreg materials tend to be a labor intensive operation associated with a lot of material handling. Campbell [3] point out that ply cutting and collation are major cost drivers that can account for 40–60% of the manufacturing cost. Ward et al. [4] note that automated solutions, that can replace manual prepreg layup, have been around since the 80's and that automated systems for dry fiber fabrics descend from the 1950's. Recently early concepts for automated composite handling, so called pick-and-place systems, have been revisited. However, Ward et al. [4] suggest that many of these attempts have been unsuccessful due to a combination of high capital requirements and low output volumes as well as technology barriers like the lack of efficient and reliable end effectors. For automated manufacturing of composite components using prepreg, Ward et al. [4] conclude that realistically, the only commercially available alternatives are Automated Tape Laying (ATL) or Automated Fiber Placement (AFP). However, there are many parts like ribs, spars and brackets that are too small to be efficiently manufactured using ATL or AFP [5]. Geometrical features such as double curvature, tight corners and steep ramps are challenging for ATL and AFP [6] and small parts are difficult to realize due to the minimal course length [7].

Resin Transfer Molding (RTM) and vacuum injection manufacturing require a great deal of dry fiber reinforcements handling. Although these manufacturing processes are sparsely used in aerospace applications they seem to be more established in the automotive sector and for manufacturing of wind turbine blades. Much like prepreg, there seem to be a lack of automated applications for the handling of dry fibers installed in full scale production. Reinhart and Ehinger [8] note that handling and draping of dry fibers to manufacture preforms is still accomplished manually. Dry fiber placement technology, is a growing area with several ongoing research projects [9]. But as for prepreg ATL and AFP there is a need for alternative solutions also for dry fiber placement when it comes to small and complex shapes. Looking at both prepreg and dry fiber based manufacturing Brecher et al. [10] note that "Manufacturing of continuous fiber-reinforced components has so far, except for specialized solutions, been a manual process".

* Corresponding author.
E-mail addresses: andreas.bjornsson@liu.se (A. Björnsson), marie.jonsson@swerea.se (M. Jonsson), kerstin.johansen@liu.se (K. Johansen).

https://doi.org/10.1016/j.rcim.2017.12.003
Received 10 May 2017; Received in revised form 9 October 2017; Accepted 21 December 2017
0736-5845/© 2017 Elsevier Ltd. All rights reserved.

Table 1
A mapping of the reviewed sources into material categories.

Dry fibers	Prepreg	Thermoplastic	Auxiliary
[8,10–12,14,18,20–36]	[13,22,23,37–42]	[10,43,44–46]	[18,19]

With increased use of FRP there is a growing need to further develop automated manufacturing technologies that cover the gaps where methods like ATL, AFP and automated dry fiber placement are unsuitable. This review paper focus on pick-and-place concepts as a way to automate material handling in composite manufacturing. These concepts can be used for common tasks like clearing the cutter table and layup of flat laminates or layup directly onto contoured molds and, as will be shown, for a wide array of materials. The aim is to highlight the challenges associated with automated composite handling and to analyze the main design principles that have been employed for such systems in terms of handling strategy, reconfigurability, gripping technology and distribution of gripping points etc. An extensive review focused on pick and place solutions of different composite related materials is lacking although several smaller compilations can be found as part of background descriptions for a particular system as for example in [12,15,18]. Also, although the review in [39] covers many of the processes linked to automated composite manufacturing, it can be too broad for the researcher or engineer looking to develop pick and place automation.

This paper reviews a wide range of academic papers, datasheets, product presentations etc. that deal with automated handling of materials used in composite component manufacturing. Most of the papers present solutions for the aerospace industry but there are also descriptions of systems aimed at the automotive sector [11–13] as well as towards the manufacturing of wind turbines [14]. Some of the reviewed papers describe different generations of the same system and these papers have been included if they add something new to the analysis, like stating a challenge, or a technical detail not covered before. Some papers describe several different systems, and thus may show up in different categories. Therefore, direct quantitative comparisons between categories in the tables cannot be made. The presentation of references in the tables is intended as a guide for the reader on where to find more information.

2. Handled materials

The term “handling” in manufacturing of composite components implies a very broad range of materials. In the review four main groups emerge; prepreg, dry fiber reinforcements, thermoplastics and auxiliary materials. Apart from papers covering these four groups some papers with more general definitions such as generally limp materials [15] and papers that do not specify exactly the type of composite material the system is designed for [16,17] are also included in the review. Papers that address several different types of materials are found in multiple categories in Table 1. The prepreg-category include both uni-directional (UD) and woven carbon fiber prepreg. For dry fibers different kinds of reinforcements such as woven or UD non-crimp fabrics and materials like carbon fibers and glass fibers are considered. The thermoplastic category includes thermoplastic-infused UD tape and materials just referred to as “thermoplastic composites” in the reviewed papers. The category auxiliary materials include thin and limp materials that are needed in the manufacturing process like tailored membranes for vacuum infusion [18] as well as peel plies, release film and breather materials [19]. The review shows that dry fibers has received a high interest and that auxiliary materials seem a novel, albeit challenging, field for automated handling.

3. Challenges associated with automated handling

Many of the reviewed sources account for challenges that must be overcome in order to design a successful system for automated handling of composite materials. The described challenges can be arranged in four categories based on what aspects they cover. The four categories are quality, material, product and system aspects. Table 2 shows what sources cover challenges in each category and below follows a more detailed description of the challenges included in each category. Many of the sources cover numerous challenges and subsequently some sources are cited in several categories.

The quality category includes general challenges posed by the high quality level that many of the systems are designed for. These are usually connected to the strict requirements employed in the aerospace industry. The category also includes more specific challenges related to requirements for contamination-free handling [18,44], positional accuracy in general [8,14,18,20,27,32,37–39] and with specified tolerances for gap/overlap [8,18,20] as well as correct fiber angles after laydown [14,18,21,27] and requirements for damage-free handling [14,26,37], for example, to avoid fiber buckling, distortion and other deformations [14].

The material category includes challenges directly related to the properties of the material that the system is designed to handle. For both prepreg and dry fiber reinforcements the limp material characteristics with highly anisotropic properties [8,24–27,40] and low structural rigidity [12,15,24–27,31,37–40] are frequently acknowledged. The permeable nature of dry fiber reinforcement is also commonly listed as a challenge [8,12,24–27,31]. For prepreg, the dominant challenge is the tack (i.e., the stickiness of the material) [38–40]. The fact that material properties are affected by ambient temperature [15,40] and that protective papers or foils are used for prepreg protection are also mentioned [40]. For thermoplastic materials the high stiffness and rigidity of the material and the subsequent need to maintain material tension in order to avoid distortion is highlighted [10,45].

The product category includes many of the challenges that originate from the shape of the final product, such as the geometry of the mold that the material must be placed on [12,18,19,24,27,38,40,44] and the multitude of different shapes and sizes [8,12,18,24–27,32,45] of material plies that are required to manufacture the product.

Some papers describe challenges associated with the production system that the handling system is part of and that it must interact with. These aspects have been mapped in the system category and it contains aspects such as complexity, weight and power consumption of the end effector [18], cycle-time requirements [8,15] and process related effects on cycle times such as required hold times for powder binder to heat up and cool down [11,35]. It also includes the need to pick up plies from a cutter surface without disturbing adjacent material [15,24] and the need for the handling system to store cut plies to improve nesting efficiency in order to reduce scrap in the cutting operation [45]. The category also includes how different processes affect each other, for example how a stitching-pattern for non-crimp fabrics that simplifies draping can complicate handling [29].

4. Handling strategies

The material handling operation can be done in many different ways, which are connected to the manufacturing process that the material handling system is designed for. The selected handling strategy impacts the complexity of the material handling system. For example, the handling operation can be moving the material from a flat surface and deposit it onto another flat surface (2D→2D) or the material can be draped onto the three-dimensional surface like a mold or form tool (2D→3D). A transition from a 2D to a 3D surface is challenging as it yields a more complex layup where draping strategies must be considered [18,44] and the draping has direct influence over the quality of the final product [21]. Table 3 shows what sources present work related with 2D or 3D material handling. Some papers present systems for both 2D and 3D handling and are subsequently listed in both categories. Papers that conclude that their solutions might work for very lightly curved surfaces have been assigned to the 2D category.

Table 2
Sources citing challenges in each of the four categories.

Quality	Material	Product	System
[8,14,15,18–21,24,26,27,32,37–39,44]	[8,10,12,14,15,21,24–27,31,37–40]	[8,12,18,19,24–27,32,38,40,44,45]	[8,11,15,18,24,26,29,35,45]

Table 3
Sources presenting work related with 2D or 3D material handling.

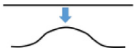
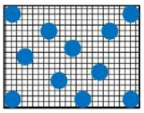
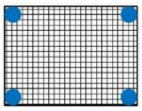
2D→2D		[13,17,22–26,32,35,37,43–45]
2D→3D		[8,10–12,16,18–22,24,27–34,36,38,40–42,46]

Table 4
Gripping technologies used for the reviewed systems.

Gipping technology	References
Vacuum	[8,11–13,16–18,20–27,32–37,40–45]
Needle	[10,19,23,28,29]
Mechanical clamp	[18,38,41]
Cryo gripper	[30,31]

Table 5
Grip point distribution.

Grip point distribution	References
Spread over the surface 	[8,11–13,16–18,22–27,32–37,42–45]
Edges or corners 	[10,18–21,28–31,38,40,41]

In the 2D case there are examples of systems designed to handle dry fibers [22–26,32,35], prepreg [13,22,23,37] and thermoplastic materials [43–45]. The 3D category contains systems for automated handling of dry fibers [8,10–12,18,20,21,24,27–34,36], prepreg [22,38,40–42] and auxiliary materials [18,19] as well as thermoplastic material [46]. In the 2D→3D category two cases [36,46] stand out from the others as the systems are designed to place many small patches of material onto a three-dimensional surface rather than draping larger plies.

5. Gripping technology

The solutions presented in the reviewed sources can also be divided based on what type of technology is used in order to achieve the grip force required to pick up and handle the material. Several of the reviewed academic papers test and evaluate different gripping technologies, but it is only the technology used in the complete end effector that are accounted for in Table 4. For the gripper type to be listed, it has to be clearly stated what technology has been used or it has been possible to determine it by reviewing pictures of the handling systems. The division has been made based on four different principles of achieving a grip force; vacuum, needle, mechanical clamping or cryo-grippers. Cryo-grippers are surface attraction type of grippers where the material is frozen to the gripper using a contact fluid. The vacuum gripper category includes several different types of vacuum systems such as ordinary vacuum systems where a high pressure difference and a low airflow generate the lift. It also includes high flow systems where the pressure difference is smaller and the airflow is much higher. In the high-flow systems the airflow can, for example, be generated by coanda ejectors or electrical fans. Sources that present systems using more than one gripping technology or that describe several systems using different technologies appear in multiple categories. Vacuum grippers are the most commonly used and the reasons for this could be that they are relatively cheap, easy to use from a technical perspective and offer the possibility of one sided handling. Needle grippers are units that extend needles through the material to lock it in place. These are less used than vacuum grippers which is a bit surprising since they are mainly designed for dry material handling, the most common material in the review. The authors have come across the opinion that needle grippers may disturb or damage the fiber during picking which is also mentioned in [40], and if that is a common worry that might lessen the use of needle grippers. However, the authors are not aware of any published, thorough investigation into the effects of needle grippers regarding quality. Mechanical clamps are systems where the material is held using fingers, which often require two-side access limiting the usefulness of the method. Although a “pinching” motion can be used for one-sided grasping, this has relatively limited applicability. Cryo grippers may be uncommon in the review since they require the addition of a freezing medium, for example water, to be dispersed over the surface of the material. This is for many industries, like aerospace, a contamination risk that can impact the final quality of the product.

6. Distribution of grip points

Two main approaches can be outlined when looking at how the grip points i.e., the points where gripping actuators generate lifting force, are distributed over the surface of a ply. The grip points can either be distributed across the whole surface of the ply or positioned along two or more of the edges or at the corners, as presented in Table 5. Handling systems with a low number of gripping actuators distributed over a large surface can grip some plies at the edges while other, often larger, plies are supported by grip points distributed over the ply surface [44]. These, hybrid systems, are categorized as giving support over the full surface as they are not specifically designed provide gripping at the outer edges or corners. Several of the systems that grip at the outer edges employ two or more robotic arms for the material handling operation [20,21,38,40,41]. Gripping point spread all over the surface can be beneficial to increase lifting force and thus handle limp materials in a controlled way. However, when handling many different ply shapes with the same gripper it can add complexity as grip points have to be rearranged or activated in order to grip a ply of a different shape (see chapter 7). Gripping at the edges may decrease complexity of the gripper as less points have to be rearranged or activated. From the reviewed material it is however not possible to find a clear connection between chosen grip point distribution, variation in ply shape and mold complexity.

7. Two-dimensional reconfigurability

All the reviewed systems are designed to pick up material from a flat 2D surface. In many cases this means to pick plies with different geometries from a cutting table without affecting the position of adjacent plies. In order to be able to grip different ply shapes most systems rely on the ability to activate or deactivate selected grip points in a pattern that matches the shape of the ply. In some systems it is also possible to change the relative positions of the grip points to adapt to the desired ply shape. Most of the systems with the ability to reposition the grip points are gripping at the outer edges [20,21,38,40,41] but there is also one

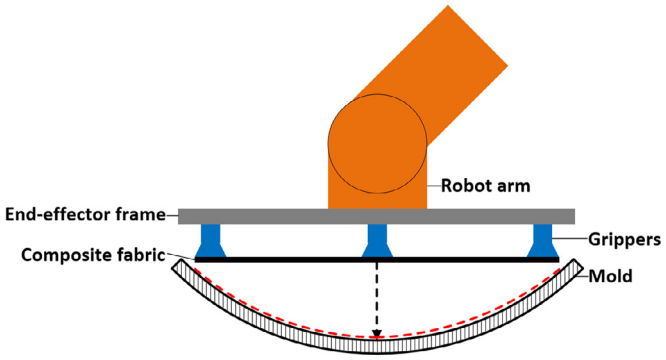


Fig. 1. Deposition of material onto a 3D mold, adapted from [31].

example [17] with the ability to reposition several grip points spread over the ply surface. Some of the systems have an ability to extend and retract the grip points, for example [21,34], but if this function is used during pickup of material from a flat surface as a way to reduce the risk of disturbing or contaminating the adjacent materials is unclear. The FORMHAND concept described in [11,12] is an example of a solution which is not capable of selectively picking a ply surrounded by other materials.

8. Three-dimensional reconfigurability

Depositing material onto a three-dimensional mold is a complex process with a lot of possible quality defects to consider, i.e., trapped air (bubbles), wrinkles, material folds etc. One of the issues that must be considered in a transition from 2D pickup to 3D deposition is illustrated in Fig. 1. For grippers that have a fixed grip of the material, the distances between the grip points must stay constant during the draping in order to avoid damaging the material or to generate wrinkles [31]. In the draping process the grip points must move relative each other, maintaining the constant distance between grippers and at the same time conforming to the mold shape. An alternative to maintaining a fixed distance during draping is to allow the material to slide along the gripper surface in order to conform to the desired shape.

The reviewed systems that are designed to deposit material onto a three-dimensional mold can be divided into four different categories based on the grip point distribution (all-over or edges/corners) and if the desired three-dimensional form is achieved actively or passively. Active draping is in this paper defined as when the desired shape (form) is achieved by grip points being purposely repositioned by actuators, the latter often part of the gripper itself (see Fig. 2).

Passive draping is defined thus as the opposite, no “activation” of grip points are done and draping is achieved for example when the end effector comes into contact with the mold and conforms to the mold shape (see Fig. 3).

Table 6 shows what sources present systems in the four categories and a more in-depth description of each category follows in the next four paragraphs.

8.1. Active draping when gripping at the edges

For active draping when gripping at outer edges or corners the majority of the reviewed systems rely on two or more robotic arms [20,21,38,40,41] and one example, the LowFlip system is shown in Fig. 4. In these cases, the relative repositioning of the grip points that is required for the draping is achieved by reprogramming the relative positions between the end effectors held by the robotic arms. There are also systems designed for a single manipulator arm and where the required repositioning of the grip points is realized internally in the end effectors [10,28,30,31] using different types of electric or pneumatic actuators. The systems presented in [20,21] rely on two arms to reposition gripping end effectors relative each other but it is also possible to adjust the positions of the grip points in the end effectors, further enhancing the draping capability of the system.

8.2. Passive draping when gripping at the edges

In [19] passive draping of auxiliary material is accomplished by using a rigid frame with grip points distributed in two rows along the sides to pick up rectangular material from a flat surface and deposit it onto a half cylindrical mold with a radius of approximately 2 m (shown in Fig. 5). The gripers are mounted onto a rotating joint to be able to be angled to the curved mold. During pick-up a rolling motion is used and during laydown the grippers are positioned a certain distance above the mold and the material is released using gravity to drape it. This drop-off procedure is necessary to avoid wrinkles since the length of the material between the gipping points is longer than mold segment it should be placed on. [19] The passive draping system described in [31] is a preface to developing an active draping system. It relies on spring loaded grippers that can pivot around a revolving joint. The grippers are mounted on a rigid frame and the system can only be used for a limited range of lightly curved mold shapes.

8.3. Active draping with grip points distributed across the surface

Gerngross and Nieberl [18] present two examples of systems for active draping. One of their systems relies on a number of discrete grip points spread over a flexible square-shaped framework. The handled

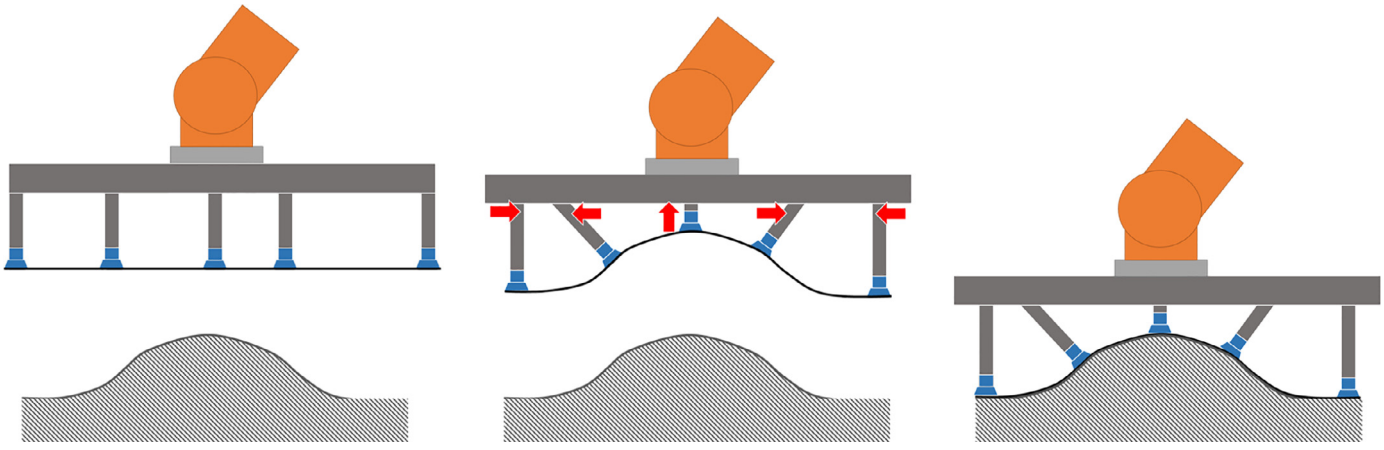


Fig. 2. A schematic description of active draping, where the gripper is repositioned purposely by actuators to form the material.

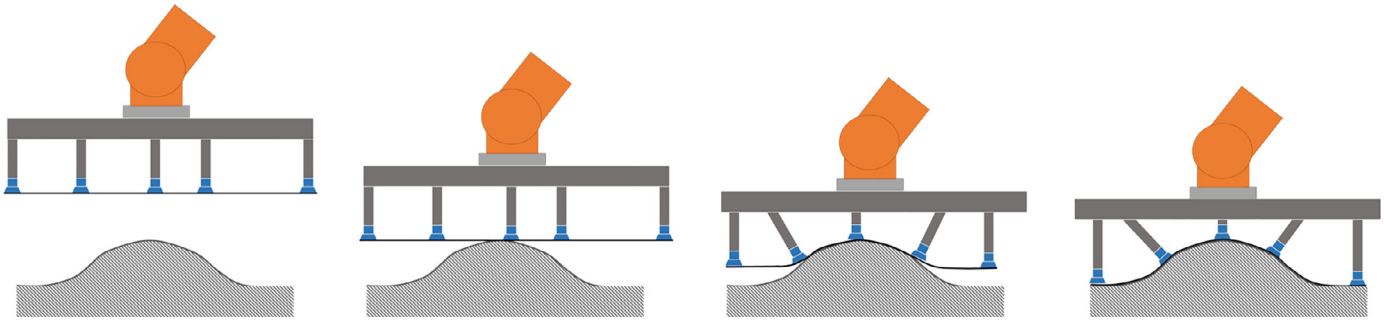


Fig. 3. A schematic description of passive draping, where draping is achieved without actuators, but for example by the gripper conforming to the surface of the tool.

Table 6
Sources describing systems with active or passive draping strategies presented based on grip point distribution.

Grip points at edges or corners		Grip points spread over surface	
Active draping	Passive draping	Active draping	Passive draping
[10,20,21,28,30,31,38,40,41]	[19,31]	[16,18,34]	[8,11,12,18,22,24,27,29,36,42,46]

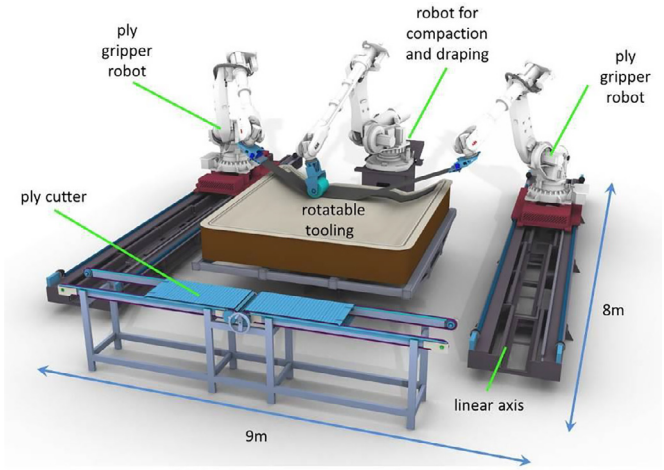


Fig. 4. Prepreg tape layup on a mold using two robots gripping at the edges and another, third, robot for simultaneous forming and compaction, making the system geometrically flexible (reprinted with permission from [41]).

material is unsupported in-between the grip points. The framework can be reshaped from a 2D shape to a double curved 3D shape by fixating the corners of the grid and apply a force at the center causing a deformation. In this system it is not possible to individually control the positions of the grip points but another system presented in the same paper relies on a high number of gripping modules whose positions can be individually controlled. The modules are assembled side by side provide support over almost the full surface of a ply and the gripper surface is transformed from a flat to a double curved shape by repositioning the gripper modules. [18] The active draping systems described in [16,34] use a matrix of grip points distributed over a large surface that are repositioned in order to achieve the desired three-dimensional shape of the handled material. All the systems with distributed grip points in combination with active draping use vacuum gripping technology.

8.4. Passive draping with grip points distributed across the surface

For the systems that rely on grip points distributed over the ply surface, passive draping is a more frequently employed strategy than active draping. Several of the systems achieve passive draping by using an end effector with an elastic outer surface that can adapt to the shape of the mold [8,22,24,27,36,46]. One example is the material handling system presented in [8,27] and illustrated in Fig. 6 which consists of a robot holding a large drum with an elastic outer layer and a large number of vacuum ports distributed over the surface. To pick up a piece of dry fiber fabric from a flat surface the robot moves the drum over the material so the drum rolls along the fiber fabric. The vacuum ports corresponding to the ply shape are gradually opened. To deposit a material onto the mold, the same type of motion is performed with a pressure towards the mold to allow the elastic outer layer to deform to the mold geometry[8,27].

End effectors that rely on an elastic outer layer are limited to geometries with low mold height i.e., low difference between the highest and the lowest point of the mold. Löchte et al. [12] conclude that grip points homogenously distributed over the material surface improve the draping performance but that existing solutions are limited to low mold height. The FORMHAND concept presented in [11,12] and shown in Fig. 7 aims to provide a solution that can be used for deeper molds. The robot-held end effector is comprised of a granulate bag whose stiffness can be altered during the handling operation. The granulate bag is set to be stiff with a flat surface to pick up dry fiber reinforcement from a flat pick-up location using vacuum. The fabric is transferred to the mold where the stiffness of the granulate bag is altered so that it becomes form-flexible. The end effector is then pushed towards the mold gradually assuming the mold geometry and draping the fabric. [11,12]

An alternative approach for passive draping that can manage more complex mold shapes is described in [35,36,46]. Instead of draping large plies the systems are designed to handle a large number of small, similarly shaped patches of dry fibers or thermoplastics that are placed on a 3D mold using a small elastic end effector.

The only system in the passive draping category with grip points distributed over the surface that use a rigid structure is described by Linke et al. [29]. The system use non-compliant grip points arranged in an arc-shape. Instead of draping the material to the mold during the drop-off sequence the system rely on a rolling motion during pick-up in order to make the material assume the mold shape. The system only work for a predefined shape, which in this case is a concave arc. [29]

The only example where a combination of active and passive draping is employed is presented by Gerngross and Nieberl [18]. The system has a small ability to adjust actively but the main draping is done by pressing a flexible foam surface onto the mold assuming the mold shape.

9. Discussion

Assuming that the reviewed sources are an accurate cross-section of the research and development conducted on handling systems for composite materials, some interesting observations can be made. One is that

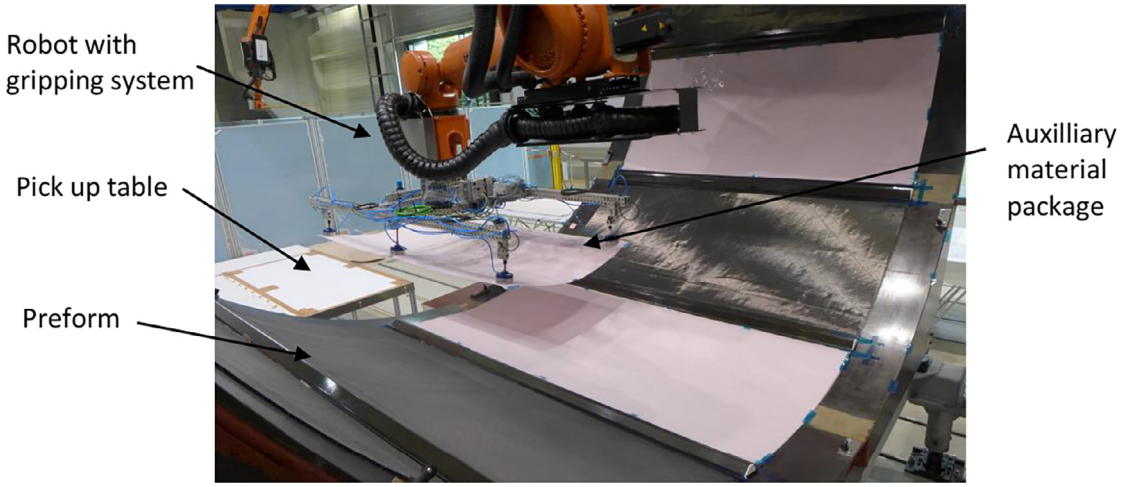


Fig. 5. Robot handling auxiliary material while gripping at the edges and employing a passive draping strategy (reprinted with permission from [19]).

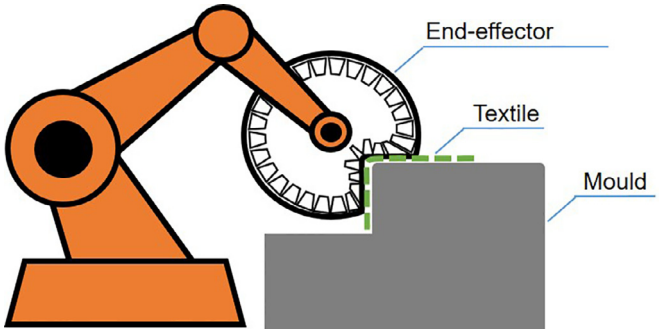


Fig. 6. Automated handling system using an elastic surface for passive draping presented in [8,27], adapted from [12].

dry fibers have received more attention than prepreg. This is despite that prepreg is the most common material for aerospace applications, where composites have been prevalent for decades. Looking at the timeline for the publications (Fig. 8), dry fibers have received a lot of attention very recently, which may be related to the increased interest from automotive, but also from aerospace. Few sources that describe systems for automated handling of auxiliary material can be found, but since materials like peel plies, breather materials and bag films are required in many composite manufacturing processes this is an interesting area to further explore, especially since the processes associated with these materials are labor intensive and are crucial to the quality of the final product.

The challenges associated with automated handling of composite materials are related to the material itself, the quality demands, the product and the system. There is a close connection between the product and the quality aspects, for example fiber angles are given by the desired properties of the product. Challenges related to quality, material and product aspects are the most cited ones. Material properties like rigidity and tack are of course central to the handling system as they impact handling sequence and end effector design. Product aspects are often linked with 3D draping, which puts extra emphasis on product geometry. Quality receives a lot of attention since many of the systems target aerospace, where quality requirements are rigorous. System aspects however should not be overlooked when designing an automated handling system. For example, end effector complexity can be reduced when designing a system for flat layup (2D→2D) followed by a subsequent forming operation compared to direct draping on a mold (2D→3D).

Although 2D → 2D systems may simplify handling, the review shows that pickup from a flat surface and layup onto a mold surface (2D→3D) is the most published strategy. This might be due to the fact that it requires a draping sequence and thereby is more technically challenging than the 2D to 2D case, making it a more interesting field of research. With exception of the FORMHAND, there is very little reported on actual implementations in industry of 2D to 3D handling. The authors can only speculate to why so few solutions has reached industrial implementation, but robustness, cycle time and equipment cost may be factors that hamper the broad use of 2D → 3D handling.

The system described in [35,36,46] where a high number of similarly shaped, small patches of material are handled, lower end effector complexity and simplifies draping (see Fig. 9). Cutting and stacking plies generate high scrap rates [47] and the approach to use a high number of

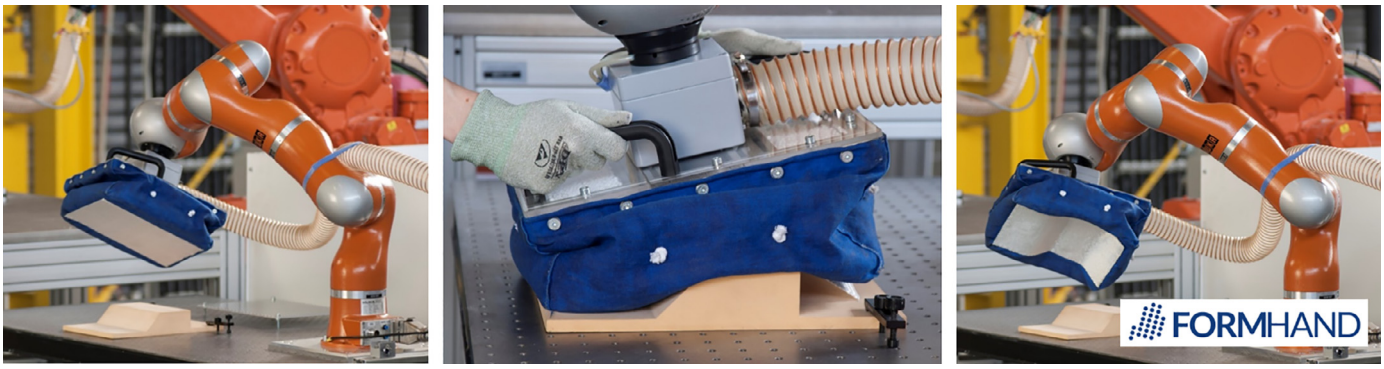


Fig. 7. The FORMHAND gripper, which uses a passive draping technique and distributed gripping (courtesy of FORMHAND, www.formhand.de).

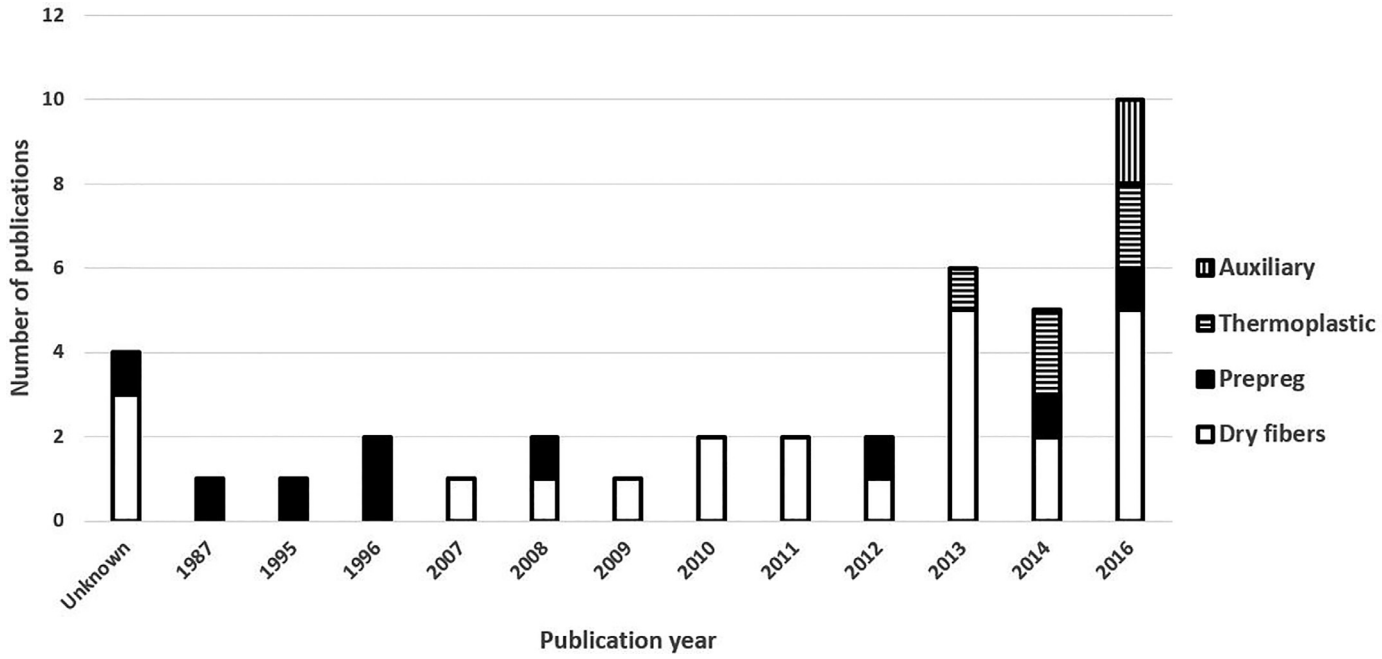


Fig. 8. Sources covering the four different material groups mapped by publication year.

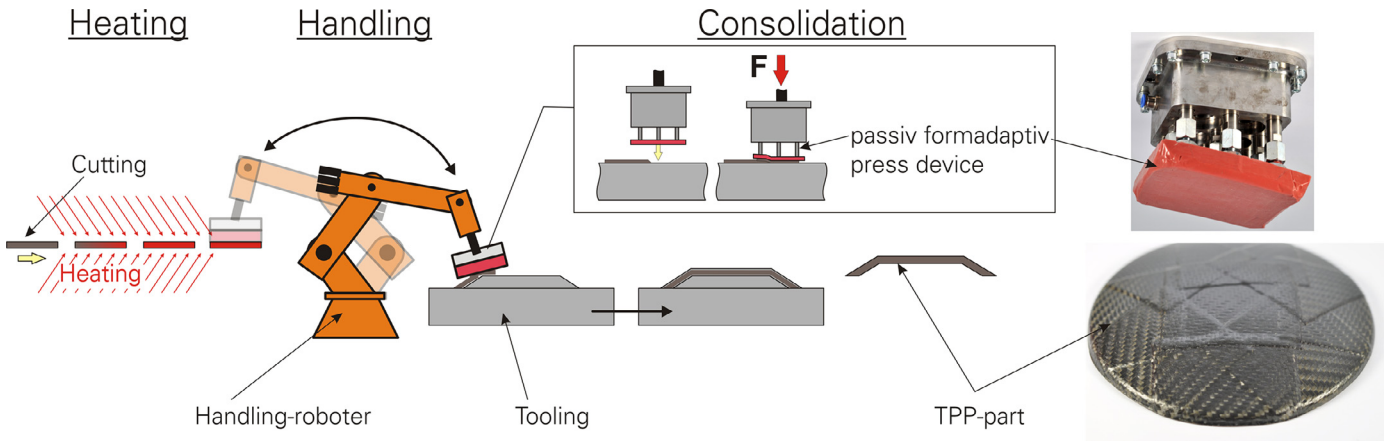


Fig. 9. The Thermoplastic patch placement (TPP) system using a passive draping approach and a simple gripper. The grippers complexity can be reduced by the use of small patches. Reprinted with permission from [46]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

small patches can improve the material utilization in the cutting process. However, cutting small patches and stacking them result in lower mechanical properties compared to using larger plies with uncut fibers. The system created by Hexcel [13], where oversized plies with equal size are stacked to a flat laminate that is later cut to the required profile allows for a simple end effector to be used. This approach can increase material scrap rates but simplifies handling and can have a great impact on the overall performance of the manufacturing system. In order to optimize nesting to reduce the scrap rate as much as possible the system clearing the cutter table presented in [45] is equipped with an additional storage drawer system that can be automatically loaded. This decouples the cutting and stacking processes so that the plies do not have to be cut in the same order as they are laid up. Active draping where the grip points in an end effector are actively repositioned using different forms of actuators drives end effector complexity and size. The use of two or more robot arms allows smaller and simpler end effector design as the robots' collective work envelope sets the limits for ply sizes to be handled. Dörsch et al. [14] point out that a higher flexibility in terms of handling different ply geometries can be achieved when using systems that do not rely on a large number of grip points distributed over the full ply surface. Passive draping systems with grip points along the edges or corners are only described in two cases [19,31] where the second case is a preface to developing a system for active draping. The approach can yield simple end effectors but on the other hand is very limited in terms of what 3D shapes can be handled which can explain the low interest for the approach. Simplifying can also produce other consequences. In Faber et al. [19] a simple, curved gripper and a sequential lay-down sequence is used to pick up and lay down material on a concave mold. Although the end effector is simpler than an active 3D forming gripper, the programmed motion to pick up and lay down a ply is more complex. Also, the pick-up sequence is a source of positional errors for the ply as it might slide during the pick-up.

Passive draping using an elastic surface struggle with height differences within the mold as there is a limit how much the elastic layer can deform. The FORMHAND concept presented in [11,12] is an interesting solution to acquire the ability to handle deeper molds but the FORMHAND, in its current state cannot pick plies that are surrounded by adjacent material which require a separate end effector for clearing the cutter table.

10. Conclusions

To conclude it can be noted that is hard to find a generic solution for automated handling of composite materials. The nature of the materials with highly varying material properties as well as the use of many different ply shapes with different fiber orientations and complex mold shapes make it difficult, and probably not desirable, to develop systems that will be able to cover all required materials and shapes. This review however have identified some trends and common technical solutions. Dry fiber materials have received increased research attention of late, and systems for 2D to 3D handling seem to be the focus. However implementations in industry is lacking, which may be due to issues regarding system robustness, cost and/or cycle time. Decoupling operations, i.e., handling from 2D to 2D and subsequently form stacked fabric, may reduce gripper complexity. Handling using distributed vacuum gripper points is preferred, but handling at the edges, mimicking a manual operator, may open up for simplified operations. Other interesting solutions to simplify handling include handling smaller pieces, using “compliant grippers” etc. Solutions for handling of auxiliary materials are almost nonexistent in research, although in aerospace such operations are both labor intensive and crucial for resulting quality.

Acknowledgements

This work was supported by the NFFP-program, funded by the Swedish Armed Forces, Swedish Defense Material Administrationand Swedish Governmental Agency for Innovation Systems.

References

[1] T. Kraus, M. Kühnel, The Global CRP-Market 2015, Carbon Composites e.V, 2015.
[2] Carbon fiber supply/demand: Outlook is bright. Composites World: Gardner Business Media Inc.; 2016. p. 1.
[3] F.C. Campbell, Manufacturing Processes For Advanced Composites, Elsevier Advanced Technology, Oxford, 2004.
[4] C. Ward, V. Bhatnagar, K. Potter, Developing an automated system for the removal of protective films from pre-preg material, to remove a manufacturing bottleneck in terms of pick and place automation, SAMPE SETEC 13, SAMPE Europe, 2013.
[5] R. Paton, Forming technology for thermoset composites, in: A.C. Long (Ed.), Composites Forming Technologies, Woodhead Publishing, 2007, pp. 239–255.
[6] M. Elkington, C. Ward, Time dependence of multi-ply forming over doubly curved shapes, ECCM17 - 17th European Conference on Composite Materials, 2016.
[7] D.H.J.A. Lukaszewicz, C. Ward, K.D. Potter, The engineering aspects of automated prepreg layup: history, present and future, Compos. Part B 43 (3) (2012) 997–1009.
[8] G. Reinhart, C. Ehinger, Novel robot-based end-effector design for an automated preforming of limb carbon fiber textiles, in: Future Trends in Production Engineering, Springer, 2013, pp. 131–142.
[9] G. Gardiner, Dry Fiber Placement: Surpassing Limits, Composites World, 2016 Out of Autoclave.
[10] C. Brecher, M. Emonts, B. Ozolin, R. Schares, Handling of preforms and prepregs for mass production of composites, in: S.V. Hoa, P. Hubert (Eds.), 19th International Conference on Composite Matateriales (ICCM19), Montreal, Canada, 2013, pp. 4085–4093.
[11] H. Kunz, C. Löchte, F. Fischer, K. Dilger, Form-flexible heating devices for integration in preform gripper, in: S.V. Hoa, P. Hubert (Eds.), 19th International Conference on Composite Materials (ICCM19), 2013.
[12] C. Löchte, H. Kunz, R. Schnurr, F. Dietrich, A. Raatz, K. Dilger, et al., Form-flexible handling technology for automated preforming, in: S.V. Hoa, P. Hubert (Eds.), 19th International Conference on Composite Matateriales (ICCM19), Montreal, Canada, 2013, pp. 4073–4084.
[13] Hexcel, Preforms for Compression Molding, 2015.
[14] C. Dörsch, H. Tummala, O. Bagemiel, Optimising high-throughput, automated preform production with non-linear simulation of the pick and place process for technical fabrics, ECCM17 - 17th European Conference on Composite Materials, 2016.
[15] G. Seliger, F. Szimmat, J. Niemeier, J. Stephan, Automated handling of non-rigid parts, CIRP Ann. 52 (1) (2003) 21–24.
[16] FibreFORM - A Complete Composite Handling System. In: Center L.T., ed. http://www.looptechnology.com/pdf/Loop%20FibreFORM%20Brochure.pdf2016.
[17] FibreMOVE - 2D Composite Gripper. In: Center L.T., ed. http://www.looptechnology.com/pdf/Loop%20FibreMOVE%20Brochure.pdf2016.
[18] T. Gerngross, D. Nieberl, Automated manufacturing of large, three-dimensional CFRP parts from dry textiles, CEAS Aeronaut. J. 7 (2) (2016) 241–257.
[19] J. Faber, C. Schmidt-Eisenlohr, Automated handling of auxiliary materials for vacuum bagging in cfrp fuselage production, ECCM17 - 17th European Conference on Composite Materials, 2016.
[20] F. Krebs, L. Larsen, G. Braun, W. Dudenhausen, Design of a multifunctional cell for aerospace CFRP production, Int. J. Adv. Manuf. Technol. 85 (1) (2016) 17–24.
[21] M. Eckardt, A. Buchheim, T. Gerngross, Investigation of an automated dry fiber preforming process for an aircraft fuselage demonstrator using collaborating robots, CEAS Aeronaut. J. 7 (3) (2016) 429–440.
[22] H. Apmann, Automatic Handling of CFRP-Material For Frame and Stringer Production, SAE, 2008 Technical Paper.
[23] H. Apmann, A. Hemmen, M. Herkt, Automatic Handling of Carbon Fiber Preforms for CFRP Parts in Aerospace, SAE, 2012 Technical Paper.
[24] G. Reinhart, G. Straβer, C. Ehinger, Highly flexible automated manufacturing of composite structures consisting of limp carbon fibre textiles, SAE Int. J. Aerosp. 2 (1) (2009) 181–187.
[25] A. Angerer, C. Ehinger, A. Hoffmann, W. Reif, G. Reinhart, G. Strasser, Automated cutting and handling of carbon fiber fabrics in aerospace industries, in: 6th Annual IEEE Conference on Automation Science and Engineering (CASE), Toronto, Canada, IEEE, 2010, pp. 861–866.
[26] G. Reinhart, G. Straßer, Flexible gripping technology for the automated handling of limp technical textiles in composites industry, Prod. Eng. 5 (3) (2011) 301–306.
[27] C. Ehinger, G. Reinhart, Robot-based automation system for the flexible preforming of single-layer cut-outs in composite industry, Prod. Eng. 8 (5) (2014) 559–565.
[28] Handling of non-rigid semi-finished textiles for FRP-production. In: IPT F., editor. Internet. http://www.ipt.fraunhofer.de/content/dam/ipt/en/documents/broschures/220_Handling%20of%20non-rigid%20semi-finished%20textiles%20for%20FRP-Production.pdf.
[29] M. Linke, C. Greb, A. Schnabel, T. Gries, Mass production technologies for textile reinforcement structures, Plast. Rubber Compos. 42 (4) (2013) 150–156.
[30] C. Greb, J. Klingele, M. Cetin, Automatisierte fertigung textiler preforms mit dem ITA-preformcenter. http://www.ita.rwth-aachen.de/1-aktuell/highlights/automatisierte%20Fertigung%20textiler%20preforms%20preforming.pdf.
[31] M.T. Kordi, M. Husing, B. Corves, Development of a multifunctional robot end-effector system for automated manufacture of textile preforms, in: 2007 IEEE/ASME international conference on advanced intelligent mechatronics, Zurich, Switzerland, IEEE, 2007, pp. 1–6.
[32] R. Reinhold, in: Solutions For Sophisticated Fiber-Composite Parts, Kunststoffe International, 2010, pp. 19–21.
[33] Integrated Composite Handling & Drape System. Broetje Automation webpage.
[34] A. Allcock, Robotic composite fibre lay-up system promises higher productivity for wing manufacture. Mach. Mag. http://www.machinery.co.uk/machinery-news/robotic-composite-fibre-lay-up-system-promises-higher-productivity-for-wing-manufacture/37654/2011.
[35] J. Snudden, B. Horn, C. Ward, K. Potter, K. Drechsler, Effects of automated patch placement on the mechanical performance of reformed NCF carbon fibre, ECCM17 - 17th European Conference on Composite Materials, 2016.
[36] F. Michl, M. Coquel, Fully-automated production of complex CFRP parts using fibre-patch-preforming technology, JEC Compos. Mag. (2014) 3.
[37] J.P. Curran, E.J. Wright, P.J. Armstrong, An intelligent vacuum gripper for robotic handling, in: Advances in Manufacturing Technology II, Springer, 1987, pp. 205–209.
[38] G. Newell, K. Khodabandehloo, Modelling flexible sheets for automatic handling and lay-up of composite components, Proc. Inst. Mech. Eng. Part B 209 (6) (1995) 423–432.
[39] G.C. Newell, R.O. Buckingham, K. Khodabandehloo, The automated manufacture of prepreg broadgoods components - a review of literature, Compos. Part A 27 (3) (1996) 211–217.
[40] R.O. Buckingham, G.C. Newell, Automating the manufacture of composite broadgoods, Compos. Part A 27 (3) (1996) 191–200.
[41] M. Szcesny, F. Heieck, P. Middendorf, R. Mezzacasa, X. Irastorza, H. Sehrschön, et al., LOWFLIP – an innovative direct 3D placement technology for plies and tapes, ECCM17 - 17th European Conference on Composite Materials, 2016.
[42] G. Gardiner, in: Automating the CH-53K’s Composite flexbeams. High-Performance Composites, Gardner Business Media Inc., Cincinnati, Ohio, 2014, pp. 74–79.
[43] M. Kühnel, A. Schuster, A. Buchheim, M. Kießig, T. Gerngross, M. Kupke, Automatisiertes endkonturnahes preforming carbonfaser-verstärkter thermoplaste mittels robotischer halbzeugablage, Deutscher Luft- und Raumfahrtkongress 2014, 2014.
[44] M. Kühnel, A. Schuster, A. Buchheim, T. Gerngross, M. Kupke, Automated near-net-shape preforming of carbon fiber reinforced thermoplastics (CFRTP), Innovative Composites Summit, JEC Europe, 2014.
[45] M. Kuehnel, A. Schuster, C. Raehtz, M. Kupke, Tailored Thermoplastic Preforming with Continuously Automated Cutting and Robotic Pick and Place Processes of Various Semi-Finished Goods, ECCM17 - 17th European Conference on Composite Materials, 2016.
[46] P. Schwanemann, N. Modler, B. Witschel, B. Grüber, F. Lenz, Process development for generative manufacturing of fiber thermoplastic composites structures – thermoplastic patch placement (TPP), ECCM17 - 17th European Conference on Composite Materials, 2016.
[47] J. Snudden, C. Ward, K. Potter, Reusing automotive composites production waste, Reinforced Plast. 58 (6) (2014) 20–27.